Negative binomial models, intro to ZIP models

Negative binomial regression

$$Y_i \sim NB(\theta,\ p_i)$$

$$\log(\mu_i) = \beta_0 + \beta_1 X_i$$

- $\mu_i = \dfrac{p_i \theta}{1 - p_i}$
- Note that θ is the same for all i
- Note that just like in Poisson regression, we model the average count
- Interpretation of βs is the same as in Poisson regression

$$Var(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

Comparing Poisson, quasi-Poisson, negative binomial

Poisson:

- Mean = λ_i
- Variance = λ_i

quasi-Poisson:

- Mean = λ_i
- Variance = $\phi\lambda_i$

negative binomial:

- Mean = μ_i
- Variance = $\mu_i + \frac{\mu_i^2}{\theta}$

Likelihood ratio test

```
m3 <- glm.nb(cigsPerDay ~ male + age + education +
               diabetes + BMI, data = smokers)
m4 <- glm.nb(cigsPerDay ~ male + age +
               diabetes + BMI, data = smokers)
m3$twologlik - m4$twologlik
```

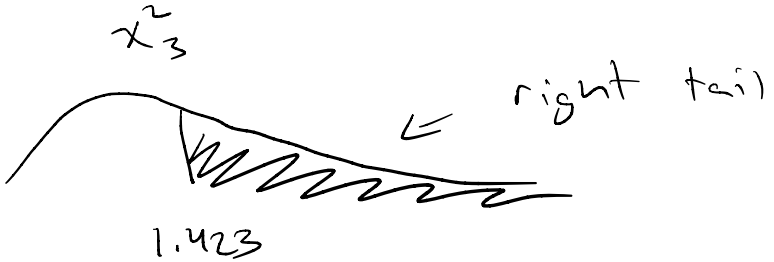
$2\log L_{full}$ $2\log L_{full}$

under H_0, $\sim \chi^2_q$ q = difference in # parameters

```
## [1] 1.423055
```

```
pchisq(1.423, df=3, lower.tail=F)
```

```
## [1] 0.7001524
```



Class activity

Work on the class activity. Submit the rendered HTML on Canvas at the end of class.

Class activity

```
m_art <- glm.nb(art ~ phd + ment + kid5, data = artic
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.233528   0.117585   1.986   0.0470 *
phd          0.008203   0.036069   0.227   0.8201
ment         0.030374   0.003240   9.373   <2e-16 ***
kid5        -0.093256   0.046475  -2.007   0.0448 *
```

(for a two-tailed test)

$H_0: \beta_{ment} = 0$

$H_A: \beta_{ment} \neq 0$

$$Z = \frac{\hat{\beta}_{ment} - 0}{SE(\hat{\beta}_{ment})} = \frac{0.0304}{0.0032} \approx 9.373$$

p-value ≈ 0

Class activity

```
m_art <- glm.nb(art ~ phd + ment + kid5, data = artic
m_art_reduced <- glm.nb(art ~ phd + kid5,
                        data = articles)
m_art$twologlik - m_art_reduced$twologlik
```

```
## [1] 77.00159
```

```
pchisq(77, df=1, lower.tail=F)
```

```
## [1] 1.70958e-18
```

p-value ≈ 0

quasi-Poisson vs. negative binomial

quasi-Poisson:

- linear relationship between mean and variance
- easy to interpret $\widehat{\phi}$
- same as Poisson regression when $\phi = 1$
- simple adjustment to estimated standard errors
- estimated coefficients same as in Poisson regression

negative binomial:

- quadratic relationship between mean and variance
- we get to use a likelihood, rather than a quasi-likelihood
- Same as Poisson regression when θ is very large and p is very small

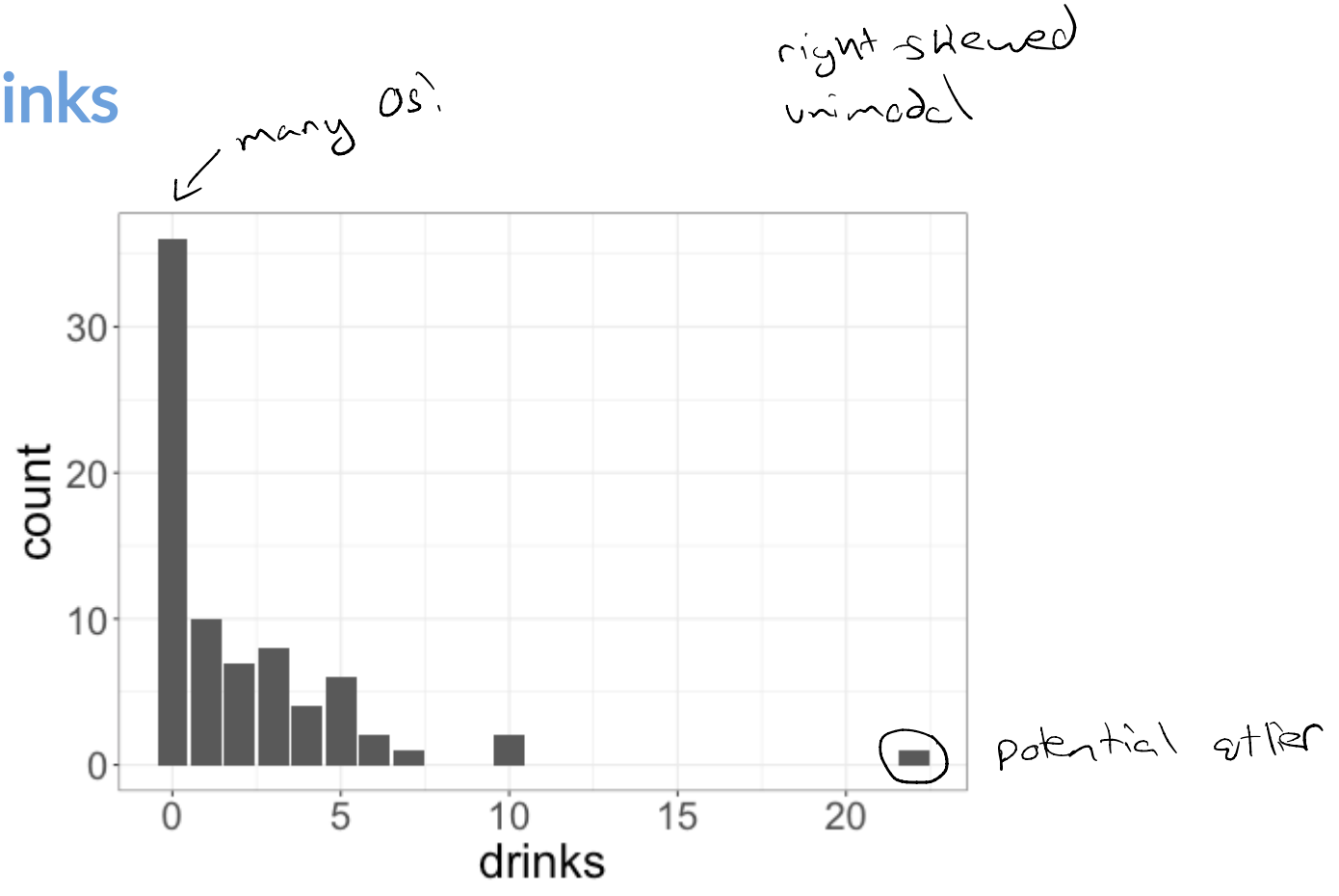
Data: College drinking

Survey data from 77 college students on a dry campus (i.e., alcohol is prohibited) in the US. Survey asks students "How many alcoholic drinks did you consume last weekend?"

- `drinks`: the number of drinks the student reports consuming
- `sex`: an indicator for whether the student identifies as male
- `OffCampus`: an indicator for whether the student lives off campus
- `FirstYear`: an indicator for whether the student is a first-year student

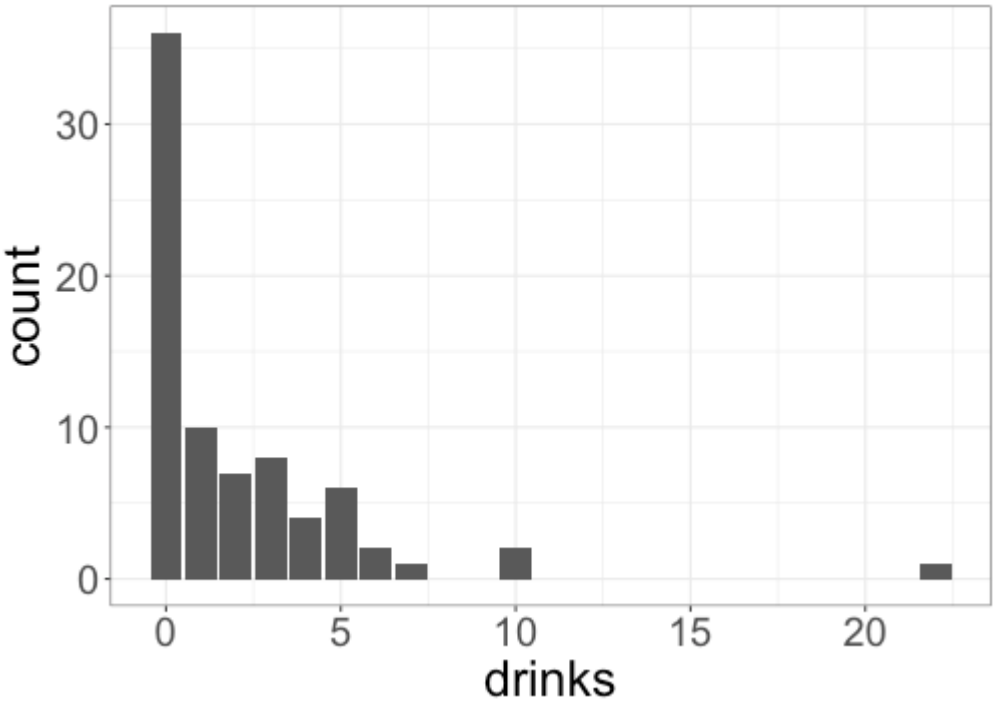
> Our goal: model the number of drinks students report consuming.

EDA: drinks



What do you notice about this distribution?

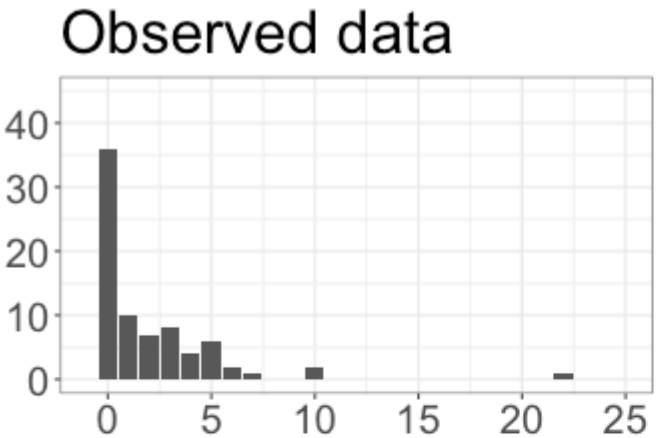
EDA: drinks

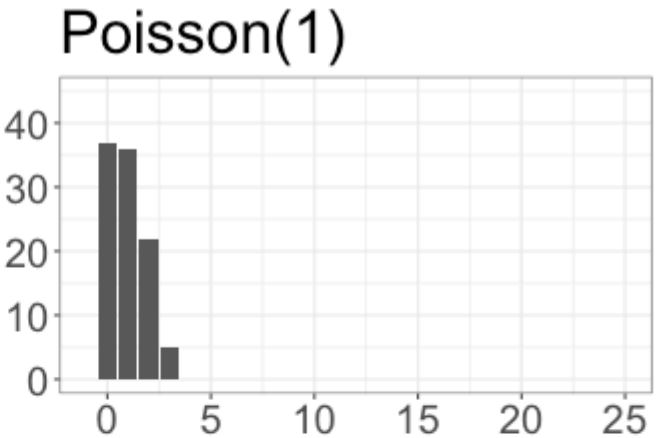


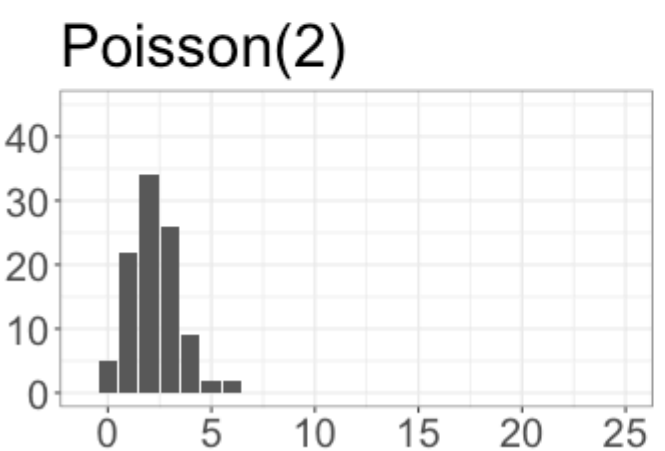
- The distribution is right skewed and unimodal
- There is an outlier near 20
- *There are more zeros than we would expect from a Poisson distribution!*

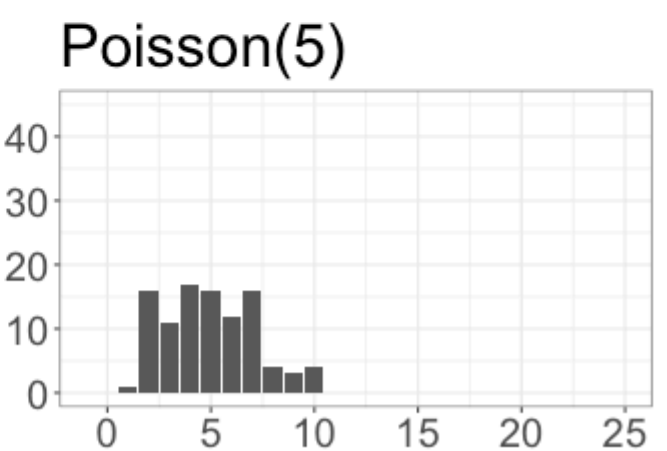
More 0s than a Poisson accounts for!

Comparisons with Poisson distributions



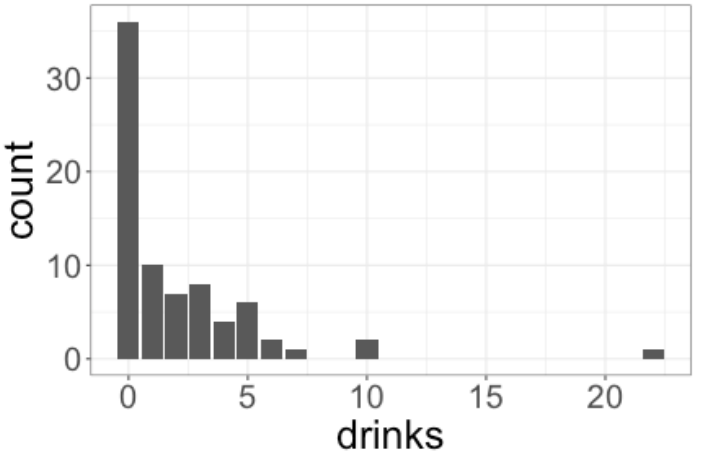






Excess zeros

> *Why might there be excess 0s in the data, and why is that a problem for modeling the number of drinks consumed?*



Excess zeros

The problem:

- There are two groups of people contributing 0s to the data: those who never drink, and those who sometimes drink but didn't drink last weekend
- By itself, a Poisson distribution doesn't do a good job modeling data that is a mixture of these two groups

> Why don't I just include whether or not the student drinks as a variable in the model?

Modeling

Let

- Z_i denote whether student i is a non-drinker (1 = never drinks, 0 = sometimes drinks)
- $\alpha_i = P(Z_i = 1)$

(Z_i is not observed in data, but we could imagine trying to model Z_i if it did exist)

We believe that α_i depends on whether or not student i is a first year.

> What model can I use for the relationship between being a first year student and being a non-drinker?

logistic regression!

Modeling non-drinkers

Z_i denote whether student i is a non-drinker (1 = never drinks, 0 = sometimes drinks)

$$Z_i \sim Bernoulli(\alpha_i)$$

$$\log\left(\frac{\alpha_i}{1-\alpha_i}\right) = \gamma_0 + \gamma_1 FirstYear_i$$

Modeling drinks

$Y_i =$ number of drinks consumed by student i

> If $Z_i = 1$ (the student never drinks), what is the probability of consuming 0 drinks?

$$P(Y_i = 0 \mid Z_i = 1) = 1$$

$$P(Y_i = 1 \mid Z_i = 1) = 0$$

$$P(Y_i = 2 \mid Z_i = 1) = 0$$

$$\vdots$$

Modeling drinks

- $Y_i =$ number of drinks consumed by student i
- Suppose that whether or not a student identifies as male and whether or not a student lives off campus has some relationship with the number of drinks consumed.

> If $Z_i = 0$ (the student sometimes drinks), how could I model Y_i ?

Poisson!

$$Y_i \mid Z_i = 0 \sim \text{Poisson}(\lambda_i)$$

$$\Rightarrow P(Y_i = y \mid Z_i = 0) = \frac{e^{-\lambda_i} \lambda_i^{y}}{y!}$$

e.g. $P(Y_i = 0 \mid Z_i = 0) = e^{-\lambda_i}$

$P(Y_i = 1 \mid Z_i = 0) = e^{-\lambda_i} \lambda_i$ etc...

So far:

$$Z_i \sim Bernoulli(\alpha_i) \qquad \log\left(\frac{\alpha_i}{1-\alpha_i}\right) = \gamma_0 + \gamma_1 FirstYear_i$$

$$P(Y_i = 0 | Z_i = 1) = 1$$

$$Y_i | Z_i = 0 \sim Poisson(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 OffCampus_i + \beta_2 Male_i$$

> Can we fit these models?

No! we don't have Z_i

idea: combine into one model (removes Z_i) try to fit combined model

Combining models

(law of total probability)

$$P(Y_i = y) = P(Y_i = y|Z_i = 0)P(Z_i = 0) + P(Y_i = y|Z_i = 1)P(Z_i = 1)$$

Case 1: $y = 0$

$$P(Y_i = 0) = \underbrace{P(Y_i = 0 \mid Z_i = 0)}_{e^{-\lambda_i}} \underbrace{P(Z_i = 0)}_{1-\alpha_i} + \underbrace{P(Y_i = 0 \mid Z_i = 1)}_{1} \underbrace{P(Z_i = 1)}_{\alpha_i}$$

$$= e^{-\lambda_i}(1-\alpha_i) + \alpha_i$$

Case 2: $y > 0$:

$$P(Y_i = y) = \underbrace{P(Y_i = y \mid Z_i = 0)}_{\frac{e^{-\lambda_i}\lambda_i^y}{y!}} \underbrace{P(Z_i = 0)}_{1-\alpha_i} + \underbrace{P(Y_i = y \mid Z_i = 1)}_{0} \underbrace{P(Z_i = 1)}_{\alpha_i}$$

$$= \frac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) \qquad y = 1, 2, 3, \ldots$$

Zero-inflated Poisson (ZIP) model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

where

$$\log\left(\frac{\alpha_i}{1-\alpha_i}\right) = \gamma_0 + \gamma_1 FirstYear_i$$

$$\log(\lambda_i) = \beta_0 + \beta_1 OffCampus_i + \beta_2 Male_i$$

> This is called a *mixture* model (it is a mixture of two different models). We *can* fit this model on the observed data (we don't need to observe Z_i)

Zero-inflated Poisson (ZIP) model

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> What do α_i and λ_i represent?

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$$\log(\lambda_i) = \beta_0 + \beta_1 OffCampus_i + \beta_2 Male_i$$

> What do α_i and λ_i represent?

α_i = probability the student doesn't drink, λ_i = average number of drinks if the student *does* drink